



Data Stream Processing

MSDS 682 (2 units)

Summer 2026

Lecture

(v2.00)

Instructor Information

Jeremy W. Gu
wgu9@usfca.edu

Office Hours

7:20-7:50pm

Office Location: Virtual Zoom

Additional Information

Right after class on Mondays

Annie Chiu
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Office Hours

N/A

Course Description

This class are to equips students with the skills necessary to process continuous data streams at scale and in real-time.

Course Overview

This hands-on course teaches students how to process and analyze real-time data streams using Apache Kafka, Python, and related data engineering tools.

Students will learn how to produce and consume Kafka messages, build streaming workflows, and interpret streaming outputs. The course also

introduces practical software engineering workflows, including Git, GitHub repositories, issues, pull requests, and project collaboration.

Python is the primary programming language. Students will use Confluent tools and Kafka Python packages for labs and projects. Each student should set up the required environment on Windows or macOS.

Prerequisite Details

Students should have basic knowledge of Python, SQL, data pipelines, statistics, and data analysis. Familiarity with pandas, data visualization, scikit-learn, and machine learning is helpful but not required.

Familiarity with the command line is useful. This course does not focus on Java, Scala, or the JVM ecosystem.

Students should use either a school or personal GitHub account. If using a personal account, students should share the account information with the instructor or TA. Students using a school email may be eligible for GitHub Education benefits.

The course includes project presentations and written reports, so clear written and spoken communication is expected.

Course Learning Outcomes

By the end of the course, students should be able to:

- Understand the Kafka ecosystem.
- Produce and consume Kafka messages using Python.
- Use Confluent tools for Kafka topic management and demos.
- Build real-time data ingestion and streaming analytics workflows.
- Interpret streaming outputs and extract insights.
- Use Git and GitHub for repositories, issues, pull requests, and collaboration.
- Use AI tools and open-source resources responsibly with proper credit.

- Present and explain technical project work clearly.

Assignments

Assignments and Final Project

There will be two assignments and one final project. Students are expected to follow all deadlines. Late submissions will not be accepted unless prior approval has been granted by the instructor.

Canvas is the official submission platform. Homework, projects, and final project materials must be submitted as ZIP files on Canvas. GitHub is used for code management, collaboration, and portfolio development.

Collaboration and Academic Integrity

Students are responsible for all submitted work. They may use AI tools, coding agents, open-source resources, online references, and other external materials to support learning, debugging, and project development. The final submission must clearly reflect the student's understanding, decisions, testing, and explanation.

Students may reuse AI-generated content, open-source code, online code, text, analysis, or other external materials, as long as the source is clearly credited and the reused parts are clearly identified. Students must provide addendum to explain what was used, which parts were assisted or reused, why the resource was used, and how the assisted or reused output performed.

Students are not allowed to submit classmates' work as their own. Collaboration is allowed for discussing concepts, techniques, and general approaches, but each student must clearly document any help received from others.

Cheating, plagiarism, or misrepresentation of work will not be tolerated and may result in penalties, including failure of the assignment or the course.

Grading Breakdown and Grading Policies

Grading Breakdown

- Attendance and Professionalism: 10%
Regular attendance, active participation, and professional conduct are expected.
- Individual Assignments: 40%
There will be two individual assignments, each worth 20%.
- Final Project: 50%
 - Project Proposal: 10%
 - Written Report and Code: 20%
 - Final Presentation: 20%

There will be no exams.

Letter Grades

This is a standard graded course with letter grades from A to F.

- A: 90–100
Excellent understanding and strong application of course concepts.
 - A+: 96–100
 - A: 93–95
 - A-: 90–92
- B: 80–89
Solid understanding and competent application of course material.
 - B+: 87–89
 - B: 83–86
 - B-: 80–82
- C: 70–79
Basic understanding of course material, with room for improvement in application and depth.
 - C+: 77–79
 - C: 73–76
 - C-: 70–72
- F: Below 70
Insufficient understanding or application of course material.

The expected class average is typically around a B or higher, depending on overall performance.

Beyond Grades

Grades reflect more than technical correctness. They also reflect professionalism, communication, collaboration, time management, and the ability to deliver quality work under deadlines.

Students are encouraged to treat this course as practical preparation for real-world data engineering and data science work. Strong performance requires technical skill, clear communication, responsible collaboration, and thoughtful decision-making.

Texts and Supplies



Kafka in Action

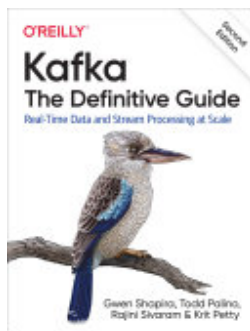
ISBN: 9781617295232

Authors: Dylan Scott, Viktor Gamov, Dave Klein

Publisher: Simon and Schuster

Publication Date: 2022-02-15

Required or recommended?: Recommended reading as a reference book.



Kafka: The Definitive Guide

ISBN: 9781492043034

Authors: Gwen Shapira, Todd Palino, Rajini

Sivaram, Krit Petty

Publisher: "O'Reilly Media, Inc."

Publication Date: 2021-11-05

Edition: 2nd Edition

Required or recommended?: Recommended. Not Required.

Course Schedule

Schedule. Each lecture: 5:30–7:20pm PDT

- **Lecture #1:** Jul 06, 2026 (M) - Zoom

- **Lecture #2:** Jul 09, 2026 (Th) - San Francisco-101 Howard 529
- **Lecture #3:** Jul 13, 2026 (M) - Zoom
- **Lecture #4:** Jul 16, 2026 (Th) - San Francisco-101 Howard 529
- **Assignment #1:** Due by 11:59pm on 7/18/2026 (20%)
- **Lecture #5:** Jul 20, 2026 (M) - Zoom
- **Lecture #6:** Jul 23, 2026 (Th) - San Francisco-101 Howard 529
- **Assignment #2:** Due by 11:59pm on 7/31/2026 (20%)
- **Lecture #7:** Jul 27, 2026 (M) - Zoom
- **Lecture #8:** Jul 30, 2026 (Th) - San Francisco-101 Howard 529
- **Final Project Proposal:** Due by 11:59pm on 8/1/2026 (10%)
- **Lecture #9:** Aug 03, 2026 (M) - Zoom
- **Lecture #10:** Aug 06, 2026 (Th) - San Francisco-101 Howard 529
- **Lecture #11:** Aug 10, 2026 (M) - Zoom
- **Lecture #12:** Aug 13, 2026 (Th) - San Francisco-101 Howard 529
- **Final Project Written Report and Code:** Due by 11:59pm on 8/14/2026 (20%)
- **Final Project Presentation:** date will be announced by a specific date (20%)

Program Learning Outcomes

Please refer to "Course Learning Outcomes" Section.

University Policies

Credit-hour Policy

One unit of credit in lecture, seminar, and discussion work approximates one hour of direct faculty instruction (or 50 minutes plus a break) and a minimum of two hours of out-of-class student work per week through one 15-week semester. For further details, see USF's [Credit Hour Policy](#).

The below resources and additional information can be found in the [Student Life Resource Toolkit](#).

Students with Disabilities

If you are a student with a disability or disabling condition, or if you think you may have a disability, contact USF Student Disability Services (SDS) within the first week of class, or immediately upon onset of disability, to speak with a disability specialist.

If you are determined eligible for reasonable accommodations, please meet with your disability specialist so they can arrange to have your accommodation letter sent to me, and we will discuss your needs for this course. For more information, please visit [Student Disability Services](#), email: sds@usfca.edu, or call (415) 422-2613.

Behavioral Expectations

The University of San Francisco is committed to providing an environment consistent with the academic nature and core values of the institution in which students can participate in learning as a humanizing, social activity rather than a competitive exercise to support the full, integral development of each person and all persons, with the belief that no individual or group may rightfully prosper at the expense of others.

It is important you know what is expected of you academically and behaviorally through the published course syllabus, the [Student Conduct Code](#), and other instructions provided by the instructor. Therefore, you are expected to uphold the following:

- Behave in accordance with the Student Conduct Code and other USF policies
- Refrain from disrupting the ability of fellow students to learn or the instructor's ability to teach. Examples of disruption include:
 - Cell phone or computer use that significantly or repeatedly distracts others
 - Coming to class late, leaving early, or excessively physically relocating oneself in the classroom
 - Speaking frequently without being called on
 - Yelling, cursing, or engaging in aggressive behavior
- When interacting online, communicate in a respectful fashion. This includes, but is not limited to:
 - Refraining from name calling, profanity, or typing in all capital letters
 - Sending multiple emails with one sentence
 - Avoiding rants or discussing non-relevant topics

Open discussion and disagreement are encouraged when done respectfully and in the spirit of academic discourse. There are also a variety of behaviors that, while not against a specific University policy, may create disruption in this course. Students whose behavior is disruptive or who fail to comply with the instructor may be dismissed from the class for the remainder of the class period and may need to meet with the instructor or Dean prior to returning to the next class period. If necessary, referrals may also be made to the Student Conduct process for violations of the Student Conduct Code.

Academic Integrity

As a Jesuit institution committed to Cura Personalis—the care and education of the whole person—USF has an obligation to embody and foster the values of honesty and integrity. All members of the USF academic community are responsible for maintaining the standards of honesty and integrity. The [honor code](#) applies to all students (undergraduate and graduate) in the College of Arts and Sciences, the School of Education, the School of Management, and the School of Nursing and Health Professions. Faculty and students in the School of Law should review their own honor code for policies and procedures. Students enrolled in distance learning (online courses) are subject to these policies as well as supplemental policies set forth by their program. All students should review and familiarize themselves with the honor code, prohibited conduct, and procedures.

Counseling and Psychological Services

Many college students experience mental health struggles. Counseling and Psychological Services (CAPS) is a great source of support for issues such as anxiety, loneliness, struggles with relationships, stress, identity development, racial/cultural concerns, and mild depression. However, CAPS does not prescribe medication and does not have a psychiatrist on staff, so students with more severe mental health concerns will be referred off-campus for treatment.

Counseling and Psychological Services (CAPS) offers remote individual and group teletherapy to students residing within California. Students seeking services are scheduled for a 15-20-minute phone triage to assess immediate risk, identify treatment needs, and provide initial recommendations. These may include a crisis intake session, brief, intermittent individual teletherapy (every 2-3 weeks), single session teletherapy, weekly individual teletherapy via UWill, weekly group therapy, or referrals to off-campus providers. There are no fees for services. To make an appointment, students must call 415.422.6352 or request an appointment via the [CAPS](#). CAPS does not accept walk-in appointments.

If you are concerned about a student and would like someone to follow up, please contact the Dean of Students Office at 415.422.5330. If you know someone who is an immediate risk of harming themselves or others please contact Public Safety at 415.422.2911 in San Francisco, or out of state dial 911, or call the National Suicide & Crisis Lifeline by dialing 988. In addition, CAPS All Hours line can be reached 24/7 by calling 855.531.076. All students are encouraged to check out [CAPS](#) and access our extensive online resources, podcasts, mental health apps, videos, self-care strategies, and more.

Title IX

The Title IX Office seeks to stop, remedy, and prevent occurrences of sex and gender-based discrimination, sexual harassment, and sexual violence. The University has a [Policy on Nondiscrimination based on Sex and Gender, Sexual Harassment and Sexual Misconduct](#). If you have experienced any of these behaviors, we encourage you to report the incident. If you report these behaviors to any staff or faculty member, they must notify the USF Title IX Coordinator.

Students who wish to report any sexual misconduct should use the [online mandatory reporting](#) form, or contact the Title IX Office directly. Other reporting options are available by visiting the Title IX [website](#). The Title IX Office is located in Lone Mountain Room 145.

As an employee at USF, and your Professor, I am a mandatory reporter, meaning I have to share any instances of sexual harassment or sexual violence shared with me or that become known to me. I will have to share this information, including names and any details known, to the Title IX Office to connect you with resources. If you would like more information about the resources available, you can ask me at any time this semester. You do not need to tell me why you are asking to get help for a friend, another student, or yourself.

Confidential Resources for Reporting Sexual Misconduct

- Students may speak to someone confidentially which will not generate a report to the Title IX Office by contacting Counseling and Psychological Services at (415) 422-6352 during M-F 9-4pm, or speaking to a clergy member in University Ministry at (415) 422-4463.
- If you need to speak to a mental health clinician immediately, please call the CAPS 24/7 All Hours Line at 855-531-0761 (available daily, including weekends and holidays, and accepts international calls), Public Safety (415-422-2911), 911, the Suicide Hotline (dial 988), or go to your nearest emergency room
- For off-campus resources, and local Bay Area organizations, view this [webpage](#).

Learning, Writing, and Speaking Centers

The Learning, Writing, and Speaking Centers (LWSC) at USF provide individualized support to assist students in better understanding course material and to aid them on their path to success. Services are free and include tutoring, collaborative peer support services, academic skills coaching, writing, and speaking support. Services are available in-person and on Zoom.

LWSC staff can be reached Monday through Thursday between 8:00am-8:00pm and Friday between 8:00am-5:00pm at LWSC@usfca.edu or through the chat box on our [myUSF webpage](#) or by phone at (415) 422-6713. To make an appointment for subject tutoring, academic skills

coaching, the writing center, or the speaking center, students should visit the [Student Appointment Dashboard](#).

Communication

All course communications, like all other USF communications, will be sent to your USF official email address. You are therefore strongly encouraged to monitor that email account.

Gleeson Library

Looking for help with a research paper or project? Set up a consultation with a Librarian or get 24/7 research help [online](#).

Attendance Policy

Mandatory attendance for every lecture.

Use of Laptops: Please keep your laptops closed for in-class lectures, unless instructed otherwise, specifically during demo or exercise sessions. This is to ensure focus and participation during lectures.

Absence Due to Illness: If you are unable to attend a lecture due to sickness or any other unavoidable circumstance, please notify me in advance.

No Distractions: Mobile phones and other electronic devices should be kept silent and should not be used during class time.

Additional USF Resources

USF Food Pantry

The USF Food Pantry is an intermediate, short-term solution for any registered USF student to receive food and toiletry resources. Students are invited to stop by the pantry located on the first floor of Gleeson Library in the Atrium, and take the items that they need. Items are available on a first-come, first-serve basis until our supply is depleted. You will be asked to check-in via QR code before entering the pantry. For more information

and the current schedule, visit the [USF food pantry website](#). If you have further questions, please contact the Pantry Coordinator at usfpantry@usfca.edu or 415-422-4099 (during business hours Monday thru Friday from 9:00am - 5:00pm). You can find out about additional food security resources through the [USF food insecurity resource page](#) and the [CalFresh resources site](#).

Use of Generative AI in This Course

The comprehensive use of generative AI tools is permitted with appropriate attribution. In alignment with the [USF Honor Code](#), students must present work that is authentically their own. In this course at the instructor's discretion, the comprehensive use of generative AI tools is permitted with appropriate attribution. As this policy may be different from other course AI policies, please ask your instructor to clarify any questions you may have. To mitigate risks associated with the use of generative AI, please review the [Guidelines for Responsible Use of Artificial Intelligence \(AI\) at USF](#).